

Design introduction:

The objective of this project is to design and fabricate a Gearbox capable of reducing a motor input speed of 11,000 RPM to a certain output speed that would be suitable to lift a 1KG mass, through a vertical distance of 1.2 Meters Within a time frame of 10s to 20s, The gearbox must achieve this while being bound to a dimension constrain of 200x200mm.

To establish this objective, we began by outlining and understanding the key technical specifications and constraints of the gearbox. Alongside this, we had to do plenty of in-depth research understanding each material and their relevant manufacturing process; all their strengths and weaknesses had to be studied and analyzed in depth. From there, multiple concept designs were brainstormed each exploring different combinations of materials, layouts, and torque transmission methods.

A Morphological Chart and Pugh Matrix were used to evaluate the generated concepts against a set of defined criteria, namely shaft strength, gear durability, ease of manufacture, cost, and compactness within the given dimensional constraints.

Brief summary of project:

This project involves the design of a model winch system used for lifting a 1 kg load over a height of 1.2 m within 10–20 seconds. This system is powered by a 2.1 W electric motor operating at 11000 rpm and uses a 3-stage reduction gearbox to reduce speed and increase output torque.

The gearbox was designed to satisfy the project constraints including a maximum stage ratio of 5:1, compact gearbox dimensions, and 8 mm shaft limitations. Engineering calculations were performed to determine the required gear ratios, gear dimensions, output speed, and torque transmission to make sure the system can safely lift the load within the required time range.

Different design ideas were compared using tools like the morphological chart and Pugh matrix to help select the most practical and efficient gearbox design for the development and CAD modelling.

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3. Summary

Task 1–Initial Design Concept

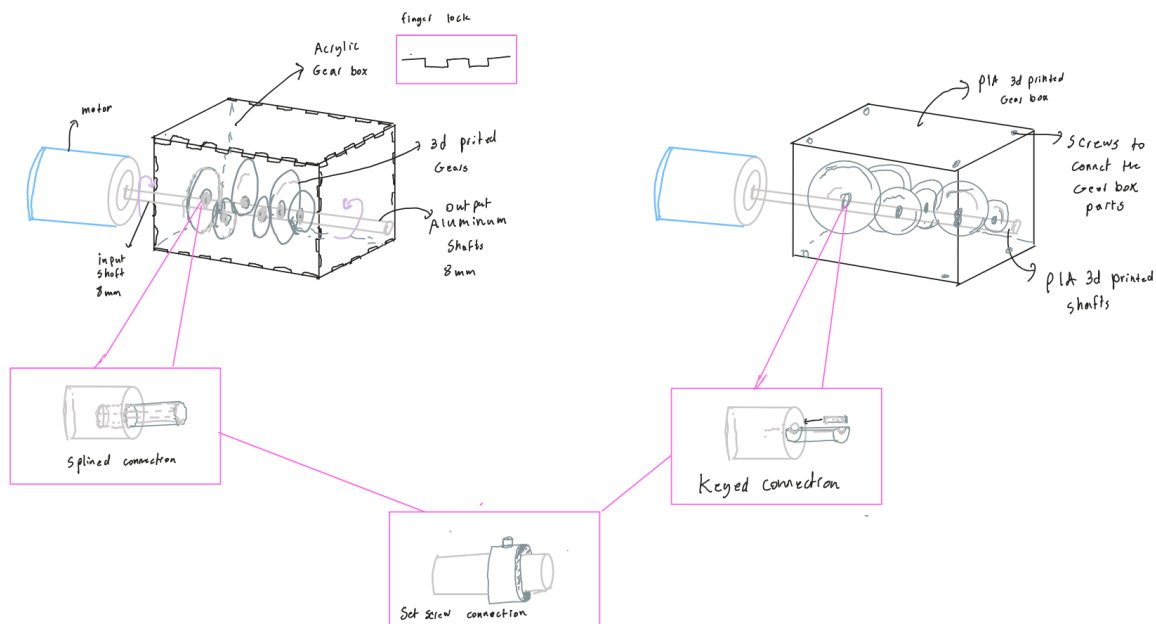
Design Considerations:

Design Consideration	Technical Specification	Importance
Load Capacity	Must lift 1 kg mass	Ensures system can handle required load safely
Lifting Height	1200 mm	Determines cable/winch travel distance
Lifting Time	Between 10–20 seconds	Prevents molten metal hazards and solidification
Motor Power	2.1 W	Limits available torque and speed
Motor Speed	11000 rpm	Determines required gear reduction ratio
Gearbox Stages	3-stage reduction system	Achieves large speed reduction efficiently
Gearbox Dimensions	Maximum 200 × 200 mm	Ensures compact gearbox housing
Shaft Diameter	8 mm at connection points	Standardizes connections and bearing compatibility
Bearing Type	608 Bearing (22 mm OD, 8 mm ID, 7 mm thick)	Supports shaft rotation and reduces friction
Input Shaft Length	50 mm protrusion	Allows motor coupling connection
Output Shaft Length	100 mm protrusion	Allows winch drum attachment
Pressure Angle	20°	Standard gear tooth geometry for smooth operation
Gear Backlash	0.1–0.2 mm	Prevents excessive friction and gear jamming
Budget Limit	250 AED maximum	Controls material and manufacturing costs
Material Selection	PLA, Acrylic, Aluminum	Affects weight, strength, manufacturability, and cost
Manufacturing Method	3D printing, laser cutting, lathe machining	Determines fabrication feasibility
Torque Transmission	Reliable gear–shaft connection required	Prevents slipping and power loss
Structural Integrity	Shaft loading and stress analysis required	Prevents shaft failure during operation
Efficiency	Minimize friction and power losses	Improves lifting performance
Safety	Controlled lifting speed	Reduces operational hazards
Durability	Lifespan estimation required	Ensures long-term reliability

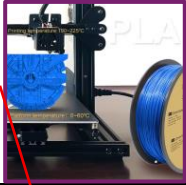
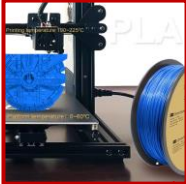
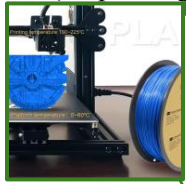
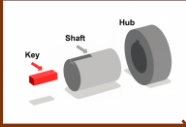
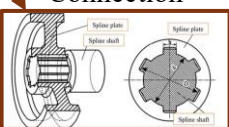
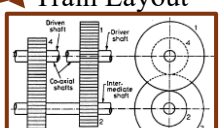
In Summary:

The winch gearbox system must be designed with goals that aim for safety and effectiveness. The system should be able to lift a 1 kg load through a height of 1200 mm within a period ranging from 10-20 seconds using a motor of an input power of 2.1 W and 11000 rpm. The gear box must include a 3-stage reduction mechanism within the dimensions of maximum 200 x 200 mm. To ensure compatibility and structural stability, standardized parts like 608 bearings and 8 mm shafts are needed. Additional considerations include gear backlash, pressure angle, material selection, manufacturing processes, cost limitations, torque transmission efficiency, and overall system durability.

2.2 Initial Brainstorming Sketches:



Morphological chart:

Design function	Option 1	Option 2	Option 3
Gears material/process	PLA (3 rd printing) 	Acrylic (Lazer cutting) 	
Shaft material/process	PLA (3 rd printing) 	Aluminum (lathe machine) 	
Gear box material/process	PLA (3 rd printing) 	Acrylic (Lazer cutting) 	
Connection type of the gear and the shaft	Keyed Connection 	Set Screw Connection 	Splined Connection 
Gears layout	Stacked Parallel Shaft 	Reverted Gear Train Layout 	

The Concepts:

Concept 1: Pla → Aluminum → Acrylic → splined → Stacked

Concept 2: Pla → Pla → Acrylic → Keyed → Reverted

Concept 3: Acrylic → Aluminum → Pla → Set screw → Stacked

Pugh chart:

Evaluation Criteria	Priority	Concept 1		Concept 2		Concept 3	
		Rating	Weighted	Rating	Weighted	Rating	Weighted
Shaft strength & durability	5	5	25	3	15	4	20
Gear ease of manufacture	5	4	20	4	20	3	15
Gear durability	5	4	20	4	20	3	15
Torque transmission reliability	4	4	16	4	16	3	12
Cost (higher = cheaper)	4	3	12	5	20	3	12
Compactness of layout	3	4	12	3	9	4	12
Assembly simplicity	3	5	15	4	12	3	9
Housing durability	2	3	6	3	6	4	8
Efficiency / Power Loss	4	5	20	3	12	4	16
Speed Reduction Performance	5	5	25	4	20	4	15
Maintenance / Repairability	3	4	12	3	9	4	12
Stability	3	5	15	3	9	4	12
Total Score			198		168		158

In Conclusion:

Based on the Pugh matrix evaluation. Concept 1 achieved the highest score overall with 198 points, making it the most reliable design. Therefore concept 1 has been selected as the final design and it will be developed and evaluated further in the next steps of the project.

Task 2 – Gear / Pinion Design Calculations:

Required Output Speed:

Given

- Lift height = 1.2 m
- Required lifting time = 10–20 s
- Assumed lifting time = 15 s

Calculating Lifting Velocity

Using:

$$v = \frac{h}{t}$$

Substituting values:

$$v = \frac{1.2}{15}$$

$$v = 0.08 \text{ m/s}$$

Converting to Drum RPM

Drum Radius Assumption

The drum radius was selected within the project constraint range:

$$r = 12 \text{ mm} = 0.012 \text{ m}$$

Angular Velocity

Using:

$$\omega = \frac{v}{r}$$

Substituting values:

$$\omega = \frac{0.08}{0.012}$$
$$\omega = 6.67 \text{ rad/s}$$

Convert Angular Velocity to RPM

Using:

$$N = \frac{\omega \times 60}{2\pi}$$

Substituting values:

$$N = \frac{6.67 \times 60}{2\pi}$$
$$N \approx 63.7 \text{ rpm}$$

Result

The output shaft should rotate at approximately:

64 rpm

Gear Ratio Calculation:

Given

- Motor speed = 11000 rpm
- Required output speed = 64 rpm

Using:

$$GR = \frac{N_{motor}}{N_{output}}$$

Substituting values:

$$GR = \frac{11000}{64}$$
$$GR \approx 172:1$$

Gear Ratio Constraint

The project restricts each stage ratio to a maximum of:

5: 1

Therefore, a practical gearbox ratio was selected:

$$5 \times 5 \times 5 = 125: 1$$

This value is close to the required ratio and acceptable for the design.

Splitting Into 3 Stages

Stage	Ratio
Stage 1	5:1
Stage 2	5:1
Stage 3	5:1

Gear Teeth Selection:

Using:

$$GR = \frac{T_g}{T_p}$$

Stage	Driver Teeth	Driven Teeth	Ratio
1	12	60	5
2	12	60	5
3	12	60	5

Module and Pitch Diameter Calculations:

Module Selection

A module of:

$$m = 1$$

was selected.

Pitch Diameter Calculations

Using:

$$d = mz$$

Pinion Diameter

$$d_p = 1 \times 12$$

$$d_p = 12 \text{ mm}$$

Gear Diameter

$$d_g = 1 \times 60$$

$$d_g = 60 \text{ mm}$$

Final Gear Dimensions

Gear	Number of Teeth	Pitch Diameter
Pinion	12	12 mm
Gear	60	60 mm

Estimated Gearbox Dimensions

Approximate gearbox dimensions were estimated based on:

- Gear spacing
- Shaft spacing
- Bearing clearance

Dimension	Approximate Size
Width	140 mm
Height	120 mm
Depth	80 mm

Task 3 – Force Analysis

Load Force Calculation:

Using:

$$F = mg$$

Substituting values:

$$F = 1.25 \times 9.81$$

$$F = 12.26 \text{ N}$$

Required Output Torque:

Using:

$$T = Fr$$

Substituting values:

$$T = 12.26 \times 0.012$$

$$T = 0.147 \text{ Nm}$$

Motor Torque Calculation:

Motor Angular Velocity

Using:

$$\omega = \frac{2\pi N}{60}$$

Substituting values:

$$\omega = \frac{2\pi(11000)}{60}$$

$$\omega = 1151.9 \text{ rad/s}$$

Motor Torque Calculation:

Using:

$$T = \frac{P}{\omega}$$

Substituting values:

$$T = \frac{1.82}{1151.9}$$

$$T = 0.00158 \text{ Nm}$$

Output Torque After Gearbox Reduction:

Using:

$$T_{out} = T_{motor} \times GR$$

Substituting values:

$$T_{out} = 0.00158 \times 125$$

$$T_{out} = 0.198 \text{ Nm}$$

Feasibility check:

Final Torque Comparison

Quantity	Value
Required Torque	0.147 Nm
Available Torque	0.198 Nm

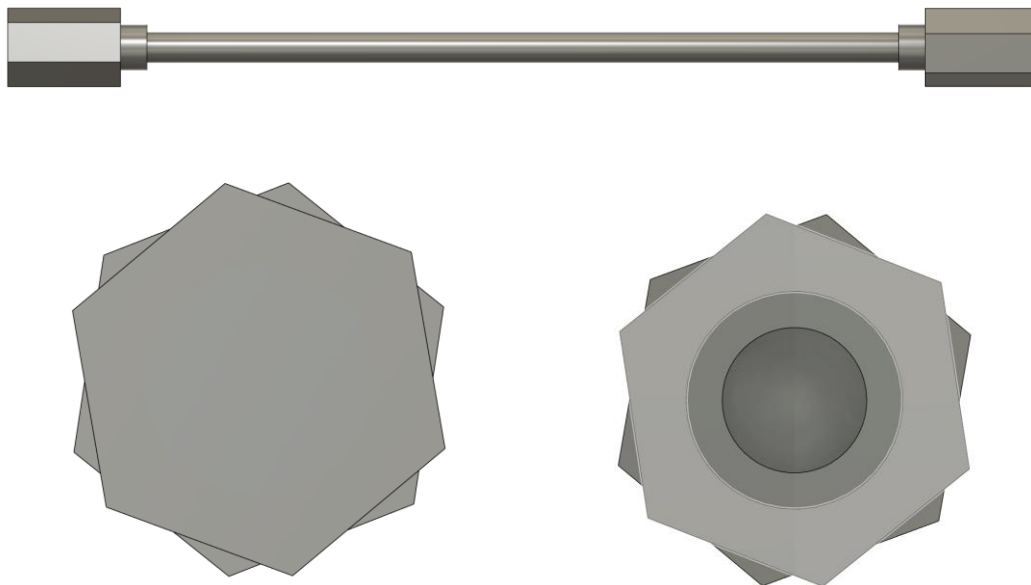
Task 4 – Shaft and Gear Design:

Overview of the splined connection:

A splined connection is a mechanical torque transmission method used to connect a rotating shaft to a gear or hub. It is made up of a number of uniformly spaced teeth or geometric shapes grooves cut into the shaft that line up with matching groove inside the gear hub. This configuration prevents rotational sliding between the connected components while enabling effective torque transfer.

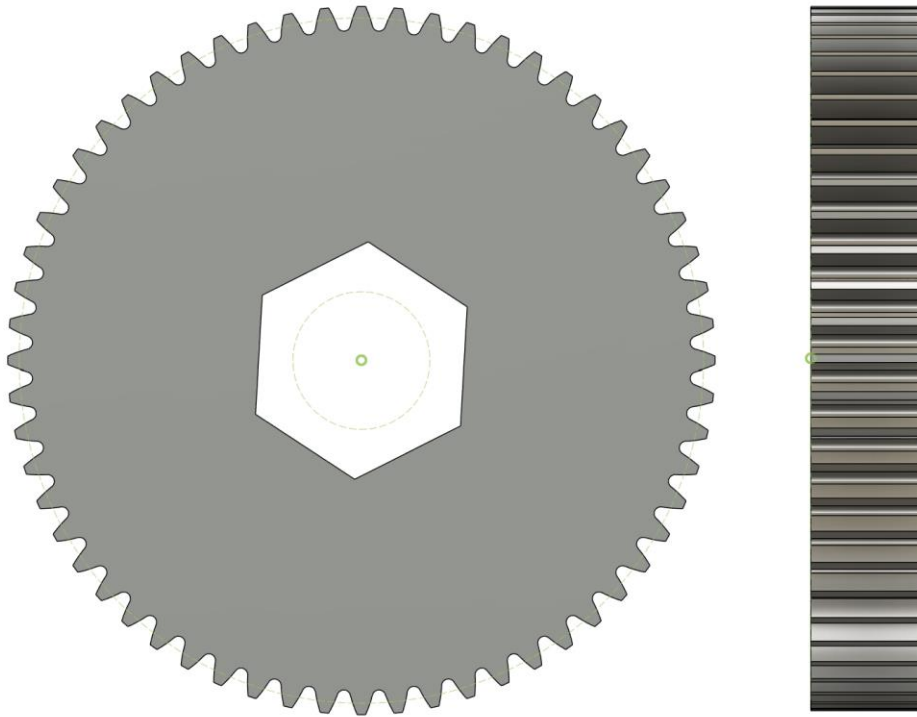
Because they offer greater torque transfer capacity, better alignment, and better load distribution than more straightforward connection techniques like set screws or keys, splined connections are frequently utilized in gearbox systems. Because of its reliability and capacity to sustain steady power transfer under constant rotating loading, the splined connection was chosen for this project.

The shaft design:



The shaft assembly was created to give the gearbox system steady rotational motion and dependable torque transfer. To increase assembly stability and support component placement, a straightforward cylindrical shaft design with stepped ends was used. For safe attachment and correct alignment on the shaft, the gear has a central bore. The overall design prioritizes efficiency in power transfer and simplicity of manufacture.

The gear design:



The gear was designed as a spur gear to provide efficient power transmission and smooth rotational motion within the gearbox system. For correct meshing and steady torque transfer between gear stages, the gear's teeth are evenly spaced. To improve shaft attachment and lessen sliding during operation, a centre hexagonal bore was added. Additionally, the design keeps its geometry simple for easy assembly and manufacturing.

The Summary and Next steps:

This project focused on the design and development of a 3-stage gearbox system for a model winch application. In order to lift a 1 kg load through a height of 1.2 m in 10–20 seconds, the gearbox had to lower the motor speed from 11000 rpm to an appropriate output speed.

Mathematical calculations were performed throughout the design process to ascertain the necessary gear ratios, torque transfer, gear size, and shaft performance while satisfying the project brief's financial and dimensional restrictions.

A morphological chart and Pugh matrix were used to generate and evaluate various design concepts. Because of their reliability, structural strength, effective torque transmission, and small gearbox arrangement, Concept 1 received the highest overall score and was chosen as the final design based on the evaluation results. CAD models of the shaft and spur gear were also developed to demonstrate the selected gearbox arrangement and support further prototype development.

The next stage of the project will focus on completing the detailed CAD assembly of the gearbox system and finalizing the positioning of the gears, shafts, bearings, and housing components. The gearbox components will be produced using the chosen techniques, such as 3D printing, laser cutting, and machining, after the CAD creation stage. The gearbox performance, output speed, torque transmission, and lifting capability under the necessary loading circumstances will next be tested on the built prototype.

Finally, the testing results will be analysed to identify possible improvements related to efficiency, durability, stability, and overall gearbox performance before completing the final project evaluation and reflection.